Science practice booklet: Grade 3 expectations

Grade 3 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 3 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: 3-LS1-1, 3-LS2-1, 3-LS3-1, 3-LS3-2, 3-LS4-1, 3-LS4-2, 3-LS4-3, 3-LS4-4

A worked example

Maryam sees a baby Arabian oryx standing next to its mother. Both have a pale coat and long, straight horns. Is the pale coat an inherited trait or a learned one? Explain.

1. An inherited trait is passed from parents to their young, like the colour of a coat.
2. The baby oryx did not have to learn to have a pale coat. It was born with it.
3. So the pale coat is an inherited trait.

Inherited: the baby is born with the coat its parents have.

Questions

1. Name the stages in the life cycle of a falcon, in order.

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2. A caterpillar grows and then changes into a butterfly. Which stage comes between the caterpillar and the adult butterfly?

   A. Egg

   B. Chrysalis (pupa)

   C. Nymph

   D. Hatchling

3. Put the life of a date palm in order: adult palm with fruit, seed, young palm, seedling.

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4. Arabian oryx live in herds. Which is a way that living in a herd helps an oryx?

   A. A herd keeps the desert cooler

   B. A herd means the oryx never need to drink

   C. Each oryx can eat more food than it could alone

   D. Many animals can watch for danger and protect the young

5. Which of these is a trait that a young camel inherits from its parents?

   A. A hump that stores fat

   B. Kneeling when its trainer says a word

   C. Walking calmly in a line of camels

   D. Knowing the way to a well

6. A falcon is trained to return to its owner's glove. Is this trait inherited or learned?

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7. Two seeds from the same date palm are planted. One gets plenty of water and sunlight, the other gets very little. Which grows taller, and what does this show?

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8. Hamdan finds the fossil of a sea shell in a rock in the UAE desert, far from the sea. What might this tell us about that place long ago?

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9. Why are a camel's wide, padded feet useful on desert sand?

   A. They store water for long journeys

   B. They make the sand cooler under its feet

   C. They spread its weight so it does not sink into soft sand

   D. They help it run faster than any other animal

10. Mangrove trees (القرم) grow in salty mud on the coast. Their roots stick up out of the mud. Why does this help the tree?

    A. The roots let the tree float on the water

    B. The roots take in air, because the mud holds very little

    C. The roots make sea water fresh for people

    D. The roots keep the tree cool

11. A small pond in a wadi dries up in a very hot summer. What might happen to the animals that lived there?

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12. Explain why the pale sandy coat of an Arabian oryx helps it survive in the desert.

Section 2: Earth and space

Standards practised: 3-ESS2-1, 3-ESS2-2, 3-ESS3-1

A worked example

The forecast for Tuesday in Dubai says 38 degrees and sunny, with no rain. Is that weather or climate? Explain.

1. Weather is what the air is like in one place on one day or over a short time.
2. Climate is the pattern of weather over many years.
3. A forecast for one day is about weather.

Weather: it describes one day.

Questions

1. What is the difference between weather and climate?

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2. Which sentence describes climate?

   A. A sandstorm hit Sharjah yesterday afternoon

   B. It is 41 degrees in Abu Dhabi today

   C. Summers in the UAE are usually hot and dry

   D. Rain is forecast for Fujairah tomorrow

3. Which tool measures how much rain has fallen?

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4. Which tool measures how hot the air is?

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5. Rainfall in millimetres from Monday to Friday was: 0, 0, 5, 12, 0. How much rain fell in the five days altogether?

   A. 17 mm

   B. 12 mm

   C. 5 mm

   D. 29 mm

6. Using the same rainfall data, on which day did it rain the most?

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7. The usual daytime temperature in one city is 22 degrees in January and 41 degrees in July. Which month is hotter, and which season is July in?

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8. Which is the best description of the climate of most of the UAE?

   A. Cold winters with heavy snow

   B. Rainy weather all year

   C. Cool and wet summers

   D. Hot summers, mild winters and very little rain

9. A sandstorm is coming while the family is at home. What is the safest thing to do?

   A. Open the windows to let the dust out

   B. Stay indoors and keep the windows and doors closed

   C. Go outside to watch the storm

   D. Play in the garden with a scarf on

10. After very heavy rain, water rushes down a dry wadi. Why should people stay away from it?

    A. Wadis are always full of animals

    B. The water is too salty to touch

    C. Fast water can carry away people and cars, even if it looks shallow

    D. Rain makes the sand too hot to walk on

11. Khalid's family visits a solar park in July, when the weather can be dangerously hot. Give two ways to stay safe from the heat.

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12. Each August for the last five years the top temperature in Abu Dhabi was above 40 degrees. What do you expect for next August? Explain.

Section 3: Matter, forces and energy

Standards practised: 3-PS2-1, 3-PS2-2, 3-PS2-3, 3-PS2-4

A worked example

Salem and Khalid pull a rope in opposite directions in a tug of war. They pull with the same strength. What happens to the rope?

1. Each boy pulls one way with a force.
2. The two forces are the same size and in opposite directions, so they are balanced.
3. Balanced forces do not change the motion of the rope, so it stays still.

The rope does not move.

Questions

1. What do we call two forces that are the same size and push or pull in opposite directions on an object?

   _____

2. In a tug of war, Team A pulls much harder than Team B. What happens to the rope?

   A. It moves towards Team A

   B. It stays still

   C. It moves towards Team B

   D. It spins round and round

3. A book rests on a table and does not move. Which two forces act on it?

   _____

4. Layla kicks a ball that is lying still. What does the kick do to the ball's motion?

   _____

5. Why does a ball that you drop fall to the ground?

   A. The ball runs out of energy

   B. Gravity, a pull from the Earth, pulls it down

   C. The air pushes it down hard

   D. Heavy things want to rest on the ground

6. A swing moves forward and back. One full swing takes 2 seconds. About how many full swings will it make in 10 seconds?

   A. 8

   B. 5

   C. 20

   D. 12

7. Omar notices that a toy car rolled down a ramp stops at about the same place each time he tries. How can he use this pattern?

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8. Hind has four things on the table. Which one will a magnet pick up?

   A. A plastic spoon

   B. A wooden pencil

   C. An aluminium can

   D. A steel paperclip

9. Can a magnet pull a paperclip without touching it? Explain.

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10. What happens when you bring the north ends of two magnets close together? What if you bring a north end and a south end together?

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11. Hind wants a cupboard door to stay shut without a lock. She fixes one magnet on the door and one on the cupboard. How should the magnets face each other?

    A. Two north ends facing each other

    B. Two south ends facing each other

    C. A north end facing a south end

    D. Only one magnet is needed, so no facing is needed

12. Two magnets keep pushing each other away when Noura tries to join them. What can she do to make them stick together? Explain.

Answer guide for parents

Section 1: Living things

1. Egg, chick, young falcon, adult falcon.
2. B. Chrysalis (pupa) The stages are egg, caterpillar, chrysalis (pupa), adult butterfly.
3. Seed, seedling, young palm, adult palm with fruit.
4. D. Many animals can watch for danger and protect the young. In a group there are more eyes to spot danger, and the adults can protect the young.
5. A. A hump that stores fat. A hump is a body part the camel is born with. The other choices are things a camel learns.
6. Learned: the falcon is taught it by its trainer.
7. The one with plenty of water and sunlight. The environment can change how a living thing grows.
8. It was once covered by the sea.
9. C. They spread its weight so it does not sink into soft sand. Wide feet spread the camel's weight over a bigger area, so it does not sink into soft sand.
10. B. The roots take in air, because the mud holds very little. The mud around the roots holds very little air, so the roots that stick up take in air.
11. Some move to find water in another place, and some may not survive. Changes in a habitat affect the animals living in it.
12. Pale colours reflect sunlight, which helps it stay cooler, and the coat matches the sand so it is harder to spot.

Section 2: Earth and space

1. Weather is what the air is like on a given day. Climate is the usual pattern of weather in a place over many years.
2. C. Summers in the UAE are usually hot and dry. Climate is a pattern over many years. The other sentences are about the weather on one day.
3. A rain gauge.
4. A thermometer.
5. A. 17 mm. Add the days with rain: 5 + 12 = 17 mm.
6. Thursday (12 mm).
7. July is hotter, and it is in summer.
8. D. Hot summers, mild winters and very little rain. The UAE has a hot, dry climate with little rain.
9. B. Stay indoors and keep the windows and doors closed. Staying indoors with the windows and doors closed keeps the dust out and keeps everyone safe.
10. C. Fast water can carry away people and cars, even if it looks shallow. Fast-moving water is strong and can sweep people and cars away.
11. Any two of: drink plenty of water, stay in the shade or indoors at midday, wear light clothes and a hat.
12. Probably above 40 degrees again, because weather patterns from past years help us predict what is likely to happen.

Section 3: Matter, forces and energy

1. Balanced forces.
2. A. It moves towards Team A. The forces are unbalanced, so the rope moves in the direction of the bigger pull, towards Team A.
3. Gravity pulls it down and the table pushes it up. They are balanced.
4. It starts the ball moving. A push is a force that can change motion.
5. B. Gravity, a pull from the Earth, pulls it down. Gravity is a force. The Earth pulls the ball downwards.
6. B. 5. 10 seconds divided by 2 seconds is 5 swings. The pattern repeats in the same time.
7. He can predict where the car will stop the next time it is rolled the same way.
8. D. A steel paperclip. Steel contains iron, so a magnet attracts it. Plastic, wood and aluminium are not attracted.
9. Yes. A magnet's force works across a small gap, so the paperclip can jump to it.
10. Two north ends push apart (repel). A north and a south end pull together (attract).
11. C. A north end facing a south end. Different ends attract, so a north end facing a south end holds the door closed.
12. Turn one magnet around. Then different ends face each other, and different ends attract.